Heterogeneous Contrastive Graph Fusion Network for Classification of Hyperspectral and LiDAR Data

Haoyu Jing, Sensen Wu, Laifu Zhang, Fanen Meng, Yiming Yan, Yuanyuan Wang, and Zhenhong Du, *Member, IEEE*

***Abstract*—In recent years, the rapid advancement of multisensory platforms has significantly increased the availability of multisource remote sensing data, facilitating its systematic application to various tasks. The joint classification of hyperspectral images (HSIs) and light detection and ranging (LiDAR) data remains a critical research topic, with a key challenge being the effective extraction and integration of complementary information from multisource remote sensing data. However, existing graph convolutional networks (GCNs)-based methods often fail to account for the heterogeneous topological relationships between HSI and LiDAR. Moreover, the discriminative power of HSI and LiDAR features extracted by existing methods is insufficient. In addition, existing methods are unable to fully exploit the rich self-supervised information present in the local neighborhood. To address these limitations, we propose a heterogeneous contrastive graph fusion network (HCGFN) for the joint classification of HSI and LiDAR data. First, we propose a branch enhancement module (BEM) to enhance the discriminative power of HSI and LiDAR. Second, a contrastive learning module (CLM) is introduced to effectively align HSI and LiDAR representations. Finally, we propose a dynamic heterogeneous graph structure learning module (DHGSLM) to model heterogeneous relationships and achieve efficient interaction and effective fusion between HSI and LiDAR. The extensive experimental results on three benchmark datasets indicate the effectiveness of the proposed HCGFN compared with other state-of-the-art methods. Specifically, under limited training samples, the proposed HCGFN outperformed state-of-the-art methods in overall accuracy (OA) by 5.10%, 2.46%, and 8.79% on datasets Trento, MUUFL, and Houston2013, respectively.**

***Index Terms*—Classification, contrastive loss, dual-stream, heterogeneous graph structure learning (GSL), hyperspectral images (HSIs), light detection and ranging (LiDAR).**

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I. Introduction

REMOTE sensing image classification plays a pivotal role in various Earth observation (EO) tasks, including environmental monitoring [1], agricultural management [2], crop mapping [3], mineral identification [4], and urban planning [5]. Hyperspectral images (HSIs), characterized by their exceptionally high spectral resolution and numerous continuous spectral bands [6], enable the precise differentiation of ground objects with similar visual appearances but distinct spectral properties [7]. Consequently, HSIs are widely utilized across multiple EO applications [8]. However, in complex scenarios, relying solely on HSI data may not effectively distinguish between ground objects with similar spectral characteristics, such as roads and rooftops made of cement [9]. This limitation underscores the challenge of obtaining comprehensive and discriminative features from a single sensor. In recent years, the rapid advancement of remote sensing technologies has led to a significant increase in the availability of multisensor data, which provides diverse perspectives on ground objects. Among these, light detection and ranging (LiDAR) has emerged as a powerful active remote sensing technology capable of capturing elevation and spatial distribution information [10], offering valuable complementary data to HSIs. As a result, the fusion of HSI and LiDAR data for enhanced feature representation and accurate ground object classification has garnered widespread research interest [11], [12].

In recent years, various methods have been explored for the joint classification of HSI and LiDAR data. Early approaches primarily relied on morphological profile (MP) [13] and attribute profile (AP) [14] techniques to extract features from HSI and LiDAR data. Following feature extraction, classical machine learning methods were employed for classification. For instance, Liao et al. [15] utilized a support vector machine (SVM) to classify features derived from MP, while Khodadadzadeh et al. [16] applied a logistic regression classifier to classify features extracted using AP. In addition, Xia et al. [17] implemented morphological opening and closing operations, combined with partial reconstruction, on the principal components of HSI and LiDAR data to obtain morphological features, which were subsequently classified using a random forest (RF) model. Although these conventional methods demonstrated promising performance at the time, they struggled to effectively capture the nonlinear relationships between different modalities, limiting their feature representation capabilities. Furthermore, these methods are highly empirical and rely on significant manual configuration.

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In recent years, the rapid advancement of deep learning techniques has significantly enhanced feature learning capabilities, leading to their widespread application in the joint classification of HSI and LiDAR data [18]. Among various deep learning approaches, convolutional neural networks (CNNs) have garnered considerable attention due to their ability to capture local image features through convolutional operations, obtain hierarchical representations, and reduce computational complexity via weight-sharing mechanisms. Several studies have explored CNN-based methods for HSI and LiDAR fusion. Chen et al. [19] used two CNNs to separately extract features from HSI and LiDAR, concatenated these features, and fed them into a fully connected (FC) network for fusion, ultimately using a logistic regression classifier to obtain the classification results. Hang et al. [20] also utilized dual CNNs for feature extraction; however, unlike Chen et al., they introduced a parameter-sharing strategy that coupled specific convolutional layers of the two CNNs. Zhao et al. [21] proposed a random walk network to enhance the spatial consistency of features extracted by CNNs. Zhang et al. [22] introduced a patch-to-patch CNN that learns hierarchical features from both modalities, providing an end-to-end classification framework. Wu et al. [11] designed a plug-and-play cross-channel reconstruction module based on a dual-branch CNN architecture, which reconstructs modality inputs to facilitate more compact feature learning. In addition, Zhao et al. [23] proposed a fractional Gabor CNN that incorporates Gabor operators to enhance the extraction of multiscale and multidirectional spatial features. Although CNN-based methods have shown promising performance, their reliance on local convolutional operations limits their ability to capture long-range semantic relationships and sequential information, often leading to a decline in performance [24].

In recent years, transformers have achieved tremendous success in the field of natural language processing due to their self-attention mechanism and powerful sequence data processing capabilities [25]. Inspired by this, vision researchers developed the vision transformer (ViT) [26], which models images as sequences of patches to capture global dependencies. However, ViT struggles with capturing local contextual semantics in remote sensing images. To address this, Hong et al. [27] proposed SpectralFormer, which learns detailed local spectral representations through groupwise spectral embedding. SpectralFormer also employs cross-layer adaptive fusion to introduce residual connections between different layers, alleviating the problem of memory degradation. Ding et al. [28] proposed a global–local transformer, which first uses CNNs to learn local spatial features from image cubes of different sizes and then employs transformers to model global spectral dependencies. Zhao et al. [29] combined CNNs and transformers into a dual-branch network to separately learn spectral and elevation features from HSI and LiDAR, and used a cross-attention mechanism for feature fusion. Xue et al. [30] introduced a hierarchical ViT that uses spatial hierarchical transformers to capture hierarchical spatial features. Yu et al. [31] developed a capsule transformer to capture global feature semantics at different contextual scales. Roy et al. [24] used LiDAR to generate classification tokens for the transformer and fused features via a cross-attention mechanism.

However, transformer-based methods struggle to capture non-Euclidean topological relationships between pixels. In contrast, graph convolutional networks (GCNs) are well-suited for handling non-Euclidean data and operate within a semi-supervised learning framework, enabling the integration of both labeled and unlabeled data [32]. GCNs have been widely applied in HSI classification [33], [34], [35], [36], and more recently, researchers have explored their use in the joint classification of HSI and LiDAR. Du et al. [37] proposed a graph fusion network that obtains joint feature inputs from HSI and LiDAR through concatenation and used k-nearest neighbors (KNNs) to construct an adjacency matrix. Guo et al. [38] introduced a dual GCN to extract spectral features from the topological graph and cosine graph of HSI and elevation features from LiDAR using a dense network. Xiu et al. [39] used a CNN encoder with an attention module to obtain fused features from HSI and LiDAR, weighted the fusion of the HSI-based adjacency matrix and the LiDAR-based adjacency matrix to obtain the adjacency matrix of the fused graph, and employed GCNs for feature propagation and aggregation. Feng et al. [40] constructed a dual-channel GCN to integrate feature representations from HSI and LiDAR and adopted a hierarchical information fusion scheme to combine features from both modalities. However, in GCNs, the aggregation weights of nodes are fixed, which limits the model's representational capability. To address this, graph attention networks (GATs) [41] were introduced to automatically learn aggregation weights during node computations. Cai et al. [42] selected tokens from HSI and LiDAR as nodes and established a series of subgraphs, ultimately using GAT to obtain embedded features of these subgraphs.

However, three critical limitations persist in existing methods. First, most existing methods adopt a dual-branch architecture to extract features from HSI and LiDAR separately. In this approach, labeled samples only contribute to the final classification loss without actively guiding the intermediate feature extraction within each branch. This will lead to insufficient discriminative power in the features extracted by the branches. Second, although contrastive learning has been introduced to enhance model representation by some existing methods, current implementations overly focus on discriminating individual nodes while ignoring the self-supervised information of local topological structures. This oversight limits the model's ability to capture deep topological patterns and complex dependencies across wider neighborhoods. Third, existing graph-based fusion methods focus only on the internal topological structures of HSI and LiDAR separately, neglecting the heterogeneous topological relationships. Moreover, they often assume that the initially constructed graph structure accurately reflects the relationships between nodes, even though this structure is generally derived from predefined rules. These predefined rules may not be optimal for the specific classification task, resulting in a mismatch between the initial graph structure and the optimal graph for the task.

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To address these challenges, we propose the heterogeneous contrastive graph fusion network (HCGFN) for the classification of HSI and LiDAR data. HCGFN integrates a dual-stream GAT architecture with a branch enhancement module (BEM) to extract discriminative features from both modalities. To further improve feature representation, we introduce a contrastive learning module (CLM) with neighborhood contrastive loss (NCL) and label-guided contrastive loss (LCL). In addition, we propose a dynamic heterogeneous graph structure learning module (DHGSLM) to adaptively learn the heterogeneous graph structure of HSI and LiDAR, enabling effective feature fusion. To sum up, the main contributions of this article are given as follows.

1) We employ a dual-stream GAT architecture to explore the internal spatial topological structures of each modality. Then, we design the BEM that utilizes label information to assist dual-stream GAT in extracting more discriminative features.

2) We introduce the CLM for alignment which comprises NCL and LCL. NCL is employed to explore the rich local topological self-supervised information in unlabeled data effectively. LCL utilizes valuable labeled information to further enhance the representation capability of the model.

3) To explicitly model the heterogeneous topological relationship between HSI and LiDAR, we propose the DHGSLM to dynamically learns the optimal heterogeneous graph structure. The DHGSLM facilitates full interaction between features of two modalities, and generates customized fusion weights for each node pair, resulting in more refined node features.

The remainder of this article is organized as follows. Section II introduces the proposed HCGFN in detail. Section III validates and analyzes the proposed method on three public datasets. Section IV concludes this article.

II. Proposed Method

In this section, the proposed HCGFN method is described. First, we provide a brief explanation of the architecture. Second, we introduce the design of PCM. Third, we carefully describe the design of dual-stream GAT. Fourth, we describe the designed CLM. Finally, we formulate the details of DHGSLM. The pipeline of our proposed HCGFN is shown in Fig. 1. First, the PCM is used to extract semantic information from HSI and LiDAR to generate superpixel nodes and graphs (see Section II-A). Then, the dual-stream GAT architecture employs two independent GAT and BEM to obtain feature representations of HSI and LiDAR (see Section II-B). Once the feature representations are obtained, the CLM uses the projection encoder to map the representations into a shared feature space and applies contrastive learning for modality alignment (see Section II-D). The DHGSLM integrates the graph structures of HSI and LiDAR, and dynamically learns the heterogeneous graph structure, thereby enabling the effective fusion of graph representations (see Section II-E).

A. Pixel Clustering Module

If the pixels are directly treated as graph nodes to construct the graph, the computational burden caused by the large adjacency and feature matrices would be unbearable, and the object-level semantic information in the data would also be overlooked. To address this, we consider clustering the pixels into superpixels and treating these superpixels as nodes to obtain the graph structure. In this article, we use the simple linear iterative clustering (SLIC) algorithm [43] to obtain the superpixels.

The initial input HSI and LiDAR are denoted as $I_H \in R^{H\times W\times C_H}$ and $I_L \in R^{H\times W\times C_L}$, respectively, where H and W represent the height and width, C_H denotes the number of spectral bands of HSI, and C_L denotes the number of channels of LiDAR. In order to reduce the impact of redundant information in HSI during superpixel computation and alleviate the computational burden, we use PCA to extract the principal components of HSI. In addition, considering that SLIC generate superpixels based on spectral and spatial similarities, it may overlook critical spatial information at ground object boundaries. Therefore, we employ the Sobel operator to extract ground object boundaries by utilizing the elevation change information contained in LiDAR. These boundary features are concatenated with HSI and LiDAR along the channel dimension to form a fused feature. We use fused features as input for superpixel segmentation helps to make the superpixel boundaries align more with the ground object contours. The formula for computing the superpixels is defined as follows:

$$\mathrm{SP} = \mathrm{SLIC}\left(\mathrm{PCA}\left(I_H\right) \| I_L \| \mathrm{Sobel}\left(I_L\right)\right) \tag{1}$$

where $\mathrm{Sobel}(I_L) \in R^{H\times W\times 1}$, $\mathrm{SP} = \{\mathrm{sp}_1, \ldots, \mathrm{sp}_n\} \in R^{N_s\times(\mathrm{N_{PCA}}+C_L+1)}$ represents a superpixel, and N_s denotes the number of superpixels, N_{PCA} denotes the number of principle components of HSI, and $\|$ denotes the concatenation operation.

After the superpixel segmentation is completed, the graph nodes of HSI and LiDAR can be obtained based on the spatial locations of the superpixels in SP. Specifically, the graph nodes of HSI are $v^H = \{v_1^H, \ldots, v_n^H \in R^{N_s\times C_H}\}$ and the graph nodes of LiDAR are $v^L = \{v_1^L, \ldots, v_n^L \in R^{N_s\times C_L}\}$, where v_i^H, and v_i^L share the same spatial location, and n is the number of nodes. The feature vector of each node is the mean of the feature vectors of the pixels it contains. Since GAT can automatically compute the aggregation weights between nodes without relying on precomputed edge weights, we define the adjacency matrix as follows:

$$A_{ij} = \begin{cases} 1, & \text{if } v_i \in \mathcal{N}_1\left(v_j\right) \text{ or } v_j \in \mathcal{N}_1\left(v_i\right) \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where $\mathcal{N}_1(v_i)$ is the 1-order neighbor set of v_i.

Through the above formulas, we obtain the adjacency matrices A^H and A^L for HSI and LiDAR, respectively. The graphs corresponding to two modalities can be represented as $G_H = (\mathcal{V}^H, A^H)$ and $G_L = (\mathcal{V}^L, A^L)$, respectively.

B. Feature Extraction Based on Dual-Stream GAT

In recent years, multiview learning has been proven to play a crucial role in learning high-level representations [44]. The core concept of multiview learning is that an object can be described from multiple perspectives, with each view containing information about a specific aspect. By integrating

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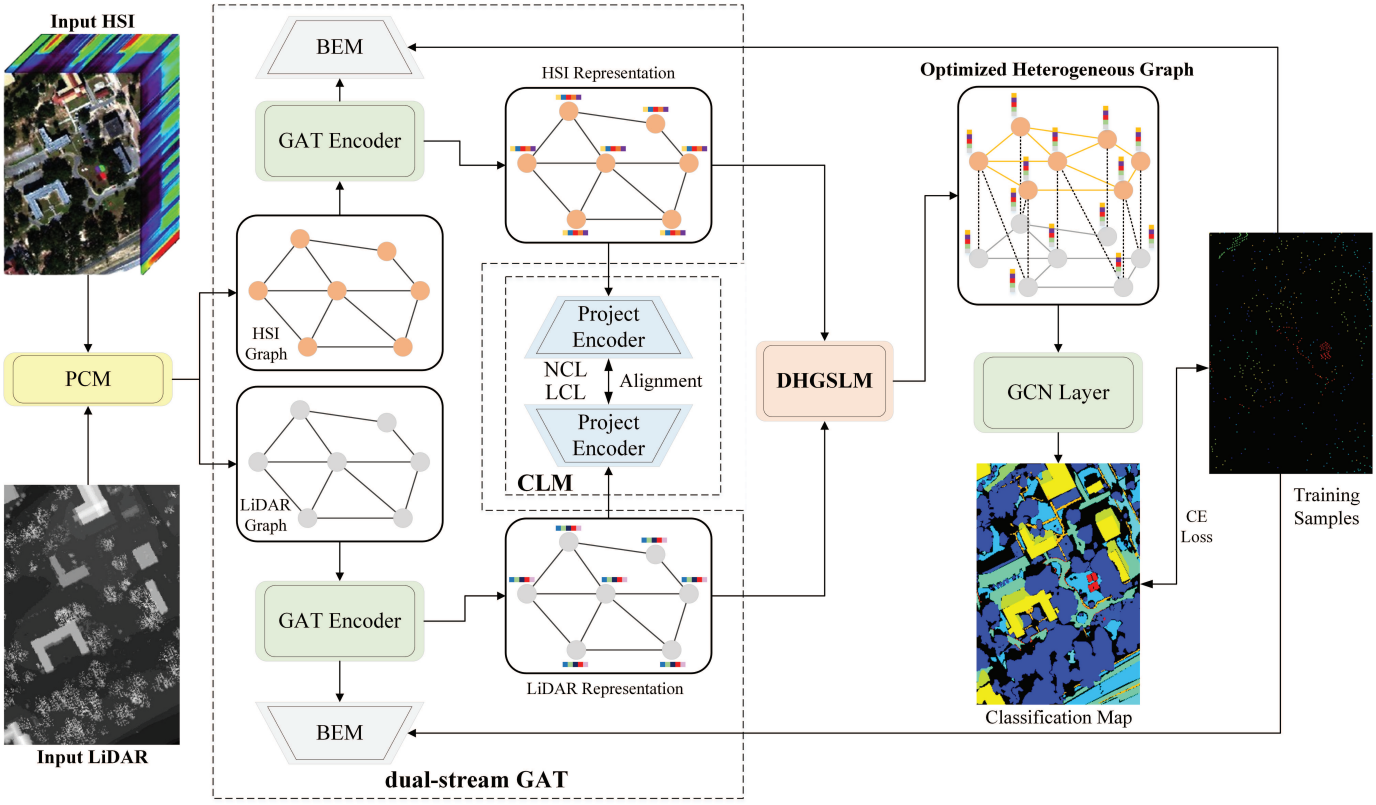


Fig. 1. Pipeline of our proposed HCGFN. The HSI and LiDAR are initially processed by the PCM to generate superpixel nodes. The graphs corresponding to HSI and LiDAR are constructed based on the topological relationships between superpixels. The dual-stream GAT assigns two GAT encoders for the HSI graph and LiDAR graph to obtain their feature representations, respectively. To enhance the discriminative power of the learned features, two BEMs are integrated into this process. Next, the modality-specific representations are further aligned through the CLM, which employs both NCL and LCL to align HSI and LiDAR representations. Finally, the aligned representations are passed to the DHGSLM. This module dynamically learns the heterogeneous graph structure of HSI and LiDAR. Finally, the refined heterogeneous graph is encoded by a GCN encoder to generate the final classification map, and the model is trained end-to-end using a cross-entropy loss function.

multiple views, a more comprehensive and rich representation of information can be obtained. Therefore, we treat HSI and LiDAR as different views of the same ground objects. To model the spatial context of HSI and LiDAR, the dual-stream GAT is designed to assign one GAT encoder to each modality, in order to obtain their respective view feature representations.

Each GAT encoder consists of a series of GAT layers, which utilize the attention mechanism to automatically capture the internal topological information and node importance within the modalities from G_H and G_L. For the GAT layer, the input is a set of node features $\boldsymbol{h} = \{h_1, h_2, \ldots, h_n\}, h_i \in R^F$ and F is the feature dimension of each node. To transform the node features into more optimal high-level features, a learnable weight matrix $W \in R^{F \times F'}$ is used to perform a linear transformation on the node features. Then, to measure the pairwise importance between nodes, we compute the attention parameter e for each adjacent node pair as follows:

$$e_{ij} = \begin{cases} a\left(h_i W \| h_j W\right), & \text{if } v_i \in \mathcal{N}_1\left(v_j\right) \text{ or } v_j \in \mathcal{N}_1\left(v_i\right) \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where $a \in R^{1 \times 2F'}$ is a learnable weight vector. Then, the LeakyReLU nonlinear activation function is applied to e, and the softmax function is used to normalize the attention

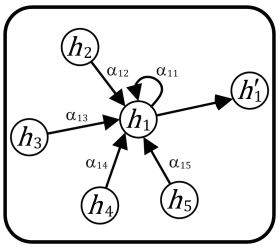


Fig. 2. Illustration of feature calculation in the GAT layer.

parameters between node v_i and all its neighboring nodes to obtain the attention coefficient α, as follows:

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(e_{ij}\right)\right)}{\sum_{k \in \mathcal{N}_1(v_i)} \exp\left(\text{LeakyReLU}\left(e_{ik}\right)\right)}. \tag{4}$$

After obtaining α, the GAT layer can use it to compute the linear combination of node features as the output high-level features, as follows:

$$h_i' = \sigma\left(\sum_{j \in \mathcal{N}_1(v_i)} \alpha_{ij} h_j W\right) \tag{5}$$

where σ is the activation function. Fig. 2 explains how to learn node importance based on the topological information

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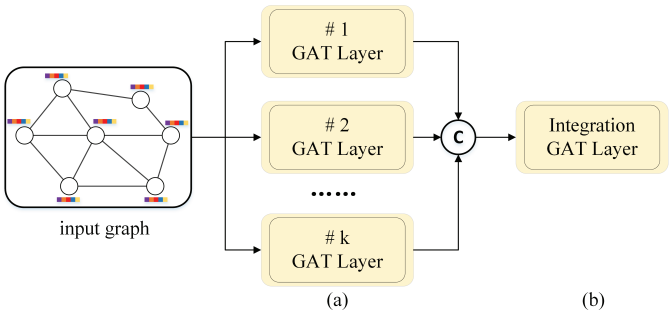


Fig. 3. Architecture of GAT encoder. It contains parallel GAT layers (a) and an integration GAT layer (b).

between adjacent node pairs. Each GAT layer is capable of autonomously gathering essential feature information from its surrounding spatial neighbors.

To enhance the model's ability to represent input features, each GAT encoder (as shown in Fig. 3) applies multiple GAT layers to simultaneously capture information at different levels. This helps comprehensively capture the diverse relationships between nodes and their neighborhoods, improving the model's robustness. After each GAT performs node feature aggregation independently, a concatenation operation is applied, as follows:

$$h_i' = \|_{k=1}^{N_{\mathrm{GAT}}} \sigma\left(\sum_{j\in\mathcal{N}_1(v_i)} \alpha_{ij}^k h_j W^k\right) \tag{6}$$

where N_{GAT} represents the number of parallel GAT layers [see Fig. 3(a)], and α_{ij}^k and W^k denote the attention coefficients and weight matrix corresponding to the kth GAT layer, respectively. To align the input feature dimensions and facilitate the subsequent construction of the joint heterogeneous graph, the GAT encoder includes an integration GAT layer [see Fig. 3(b)] to combine the features output by the multiple GAT layers, as follows:

$$z_i = \sigma\left(\sum_{j\in\mathcal{N}_1(v_i)} \alpha_{ij}^{ig} h_j' W^{ig}\right) \tag{7}$$

where $W^{ig} \in R^{(F'\times N_{\mathrm{GAT}})\times F^{ig}}$, F^{out} is the dimension of the feature output by the integration GAT layer. Finally, the output features of the GAT encoder for HSI and LiDAR can be obtained as $Z^H = \{z_1^H, \ldots, z_n^H\}$ and $Z^L = \{z_1^L, \ldots, z_n^L\}$ with N_{GAT}^H and N_{GAT}^L, respectively.

In addition, we design the BEM to fully leverage the precious labeled sample information to help the dual-stream GAT generate more discriminative features. The structure of BEM is simple that it employs a GAT layer to map the concatenated features h_i' from the GAT encoder to the classification target space, as shown by

$$h_i^{\mathrm{BEM}} = \sigma\left(\sum_{j\in\mathcal{N}_1(v_i)} \alpha_{ij}^{\mathrm{BEM}} h_j' W^{\mathrm{BEM}}\right) \tag{8}$$

where $W^{\mathrm{BEM}} \in R^{F'\times N_c}$, and N_c denotes the number of classes. Then, the h_i^{BEM} and labeled samples are used to calculate the cross-entropy loss as follows:

$$L_{\mathrm{BEM}} = -\frac{1}{N_l}\sum_{i\in yl}\sum_{j=1}^{N_c} Y_{ij}\log h_{ij}^{\mathrm{BEM}} \tag{9}$$

where N_l denotes the number of labeled nodes, Y denotes the label matrix, and yl is the set of indices of the labeled samples used for training in Y. As a result, we can obtain L_{BEM}^H and L_{BEM}^L corresponding to HSI and LiDAR, respectively.

C. Contrastive Learning Module

In cross-modal learning, alignment facilitates the acquisition of more consistent and complementary features [45]. We design the CLM for alignment which maps the feature representations of HSI and LiDAR into a shared cross-modal representation space and employs contrastive learning to optimize the feature representations. The principle of contrastive learning optimization is to minimize the feature distance for similar samples and maximize the feature distance for different samples [46], [47]. Based on this principle, we design the NCL and LCL.

Since the properties of HSI features and LiDAR features differ greatly, directly applying contrastive learning might increase the optimization difficulty in the contrastive learning process. Therefore, we designed a projection encoder to map HSI features and LiDAR features into embedding spaces of the same dimensionality for easier semantic sharing across modalities. The projection encoder contains two trainable parameters, W_1^{pe} and W_2^{pe}. The projected features can be represented as

$$x_i = \sigma\left(z_i W_1^{\mathrm{pe}}\right) W_2^{\mathrm{pe}} \tag{10}$$

where $W_1^{\mathrm{pe}} \in R^{ig\times \mathrm{CL1}}$, $W_2^{\mathrm{pe}} \in R^{\mathrm{CL1}\times \mathrm{CL2}}$, and CL1 and CL2 denote the feature project dimensions of W_1^{pe} and W_2^{pe}. The CLM provides a separate projection encoder for both HSI and LiDAR. After obtaining the corresponding projected features $X^H = \{x_1^H \ldots, x_n^H\}$ and $X^L = \{x_1^L \ldots, x_n^L\}$ from each, the NCL and LCL are computed.

1) Neighborhood Contrastive Loss: The classical multiview contrastive learning approach treats the same sample as a positive pair and different samples as negative pairs [48]. However, this approach focuses solely on node features, overlooking the local topological context, and is susceptible to noise from individual nodes. To address these limitations, we propose the NCL, which aggregates the feature information of neighboring nodes to emphasize local structural information and promote greater spatial consistency within the local feature space.

The s-order neighborhood of v_i can be denoted as

$$\mathcal{N}_s(v_i) = \mathcal{N}_{s-1}(v_i) \cup \mathcal{N}_1\left(\mathcal{N}_{s-1}(v_i)\right) \tag{11}$$

where $\mathcal{N}_0(v_i) = v_i$. In this article, we use the second-order neighborhood $\mathcal{N}_2(v_i)$ to aggregate the neighboring information of nodes [as shown in Fig. 4(a)]. We compute the second-order adjacency matrix using a radial basis function (RBF), as follows:

$$A_2[i,j] = \begin{cases} -\gamma\|v_i - v_j\|^2, & v_i \in \mathcal{N}_2(v_j) \text{ or } v_j \in \mathcal{N}_2(v_i) \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

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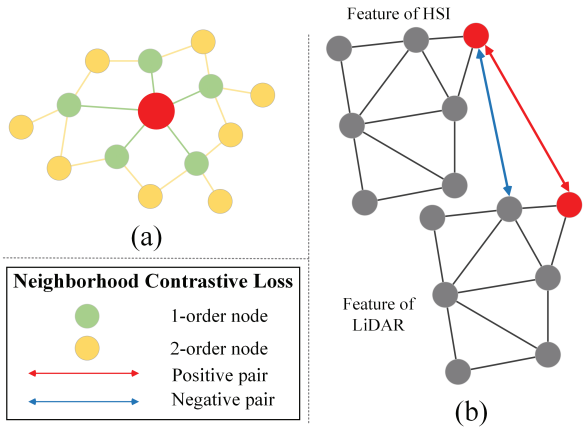


Fig. 4. Illustration of NCL.

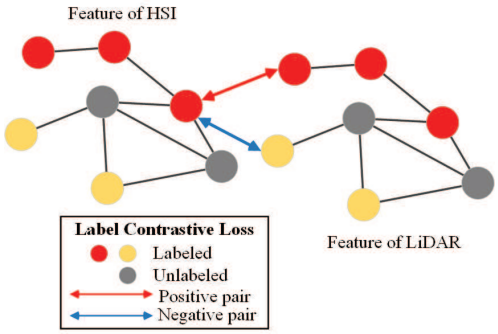


Fig. 5. Illustration of LCL.

where γ is set to 0.2. As a result, the corresponding second-order adjacency matrices for HSI and LiDAR are obtained, denoted as A_2^H and A_2^L, respectively. The node features of HSI and LiDAR after aggregating the neighborhood information can be represented as

$$\boldsymbol{X^H} = A_2^H X^H, \boldsymbol{X^L} = A_2^L X^L \tag{13}$$

where $\boldsymbol{X}^H = \{\boldsymbol{x}_1^H \ldots, \boldsymbol{x}_n^H\}$ and $\boldsymbol{X}^L = \{\boldsymbol{x}_1^L \ldots, \boldsymbol{x}_n^L\}$. The process of calculating the NCL [as shown in Fig. 4(b)] can be expressed as

$$L_{\mathrm{NCL}} = -\frac{1}{2N_s}\sum_{i=1}^{N_s}\left(\log\frac{\exp\left(\mathrm{sim}\left(\boldsymbol{x}_i^H, \boldsymbol{x}_i^L\right)/\tau\right)}{\sum_{j=1}^{N_s}\exp\left(\mathrm{sim}\left(\boldsymbol{x}_i^H, \boldsymbol{x}_j^L\right)/\tau\right)} + \log\frac{\exp\left(\mathrm{sim}\left(\boldsymbol{x}_i^L, \boldsymbol{x}_i^H\right)/\tau\right)}{\sum_{j=1}^{N_s}\exp\left(\mathrm{sim}\left(\mathbf{x}_i^L, \boldsymbol{x}_j^H\right)/\tau\right)}\right) \tag{14}$$

where τ is the temperature coefficient, and $\mathrm{sim}(\cdot)$ is the cosine similarity function.

2) Label-Guided Contrastive Loss: Classical contrastive learning methods often adopt an unsupervised individual discrimination approach, which focuses on individual differences while neglecting interclass differences. Therefore, we employ LCL to leverages the valuable labeled sample information to enhance the consistency within the same class, thus improving the interclass separability of the learned features. The process of calculating the LCL is shown in Fig. 5 and can be expressed as

$$L_{\mathrm{LCL}} = -\frac{1}{2N_l}\sum_{i=1}^{N_l}\left(\log\frac{\sum_{k=1}^{N_l}\mathbf{1}_{[y_i=y_k]}\exp\left(\mathrm{sim}\left(x_i^H x_k^L\right)/\tau\right)}{\sum_{j=1}^{N_l}\exp\left(\mathrm{sim}\left(x_i^H x_j^L\right)/\tau\right)} + \log\frac{\sum_{k=1}^{N_l}\mathbf{1}_{[y_i=y_k]}\exp\left(\mathrm{sim}\left(x_i^L, x_k^H\right)/\tau\right)}{\sum_{j=1}^{N_l}\exp\left(\mathrm{sim}\left(x_i^L, x_j^H\right)/\tau\right)}\right) \tag{15}$$

where $\mathbf{1}_{[y_i=y_k]}$ equals to 1 if the equation inside the parentheses holds, otherwise 0.

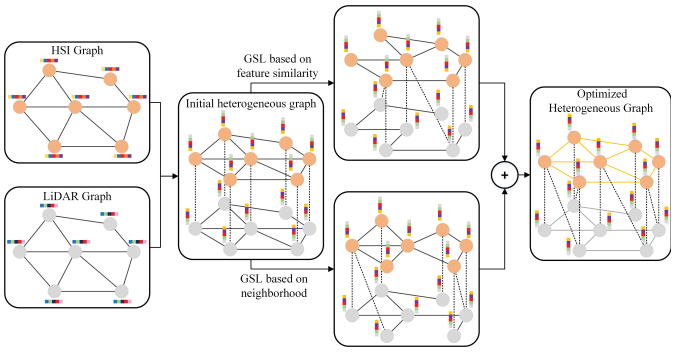


Fig. 6. Pipeline of DHGSLM.

D. Dynamic Heterogeneous GSL Module

Considering the obvious spatial correspondence between HSI and LiDAR, DHGSLM integrates the HSI graph and LiDAR graph into a heterogeneous graph. This not only allows the utilization of the feature representation information from both HSI and LiDAR but also facilitates the interaction of HSI and LiDAR modal information based on spatial topological structure, which greatly aids in complementary information between the two modalities.

It is important to note that the edges in the heterogeneous graph play a crucial role in whether good feature representations can be learned. In graph representation learning, connections between nodes with similar properties enhance the quality of feature representations [49]. Conversely, connections between nodes with significantly different properties can reduce the discriminability of the feature embeddings [50], [51], [52]. Given that the initially created heterogeneous graph may contain incorrect or missing edges [53], and that the graph includes both homogenous relationships within the same modality and complex heterogeneous interactions, DHGSLM automatically performs graph structure learning (GSL) based on the features and spatial topological relationships of HSI and LiDAR, enabling the acquisition of a heterogeneous graph structure for effective feature fusion.

Specifically, the inputs of DHGSLM are corresponding adjacency matrices A^H and A^L of HSI and LiDAR, and features Z^H and Z^L output by dual-stream GAT. The pipeline of DHGSLM is shown in Fig. 6. First, connections are added between corresponding nodes in the HSI graph and LiDAR graph. The

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initially constructed heterogeneous graph $\boldsymbol{A} \in R^{2N_s \times 2N_s}$ can be represented as follows:

$$\boldsymbol{A} = \begin{bmatrix} A^{H} & A^{H\leftrightarrow L} \\ A^{H\leftrightarrow L} & A^{L} \end{bmatrix} \tag{16}$$

where $A^{H\leftrightarrow L} = \boldsymbol{I}$, and $\boldsymbol{I}$ denotes an identity matrix that which signifies those corresponding nodes in the HSI and LiDAR graphs are connected one-to-one. Thus, it can be observed that $\mathbf{A}$ contains three types of relationships: internal connections within the HSI graph, internal connections within the LiDAR graph, and connections between the HSI and LiDAR graphs. These can be expressed as

$$\text{rel} = \{H\leftrightarrow H, L\leftrightarrow L, H\leftrightarrow L\}. \tag{17}$$

The core idea behind the optimization of $\boldsymbol{A}$ in DHGSLM is to add connections between pairs of nodes with similar feature properties, while removing connections between dissimilar nodes. In addition, to avoid the graph structure becoming too sparse and leading to insufficient spatial context information for the nodes, we also decide to diffuse the node similarity information to the neighborhood scale. Thus, DHGSLM consists of two key components: feature-based GSL and neighborhood-based GSL.

1) Feature-Based GSL: A natural approach is to utilize node feature information to learn a more optimal graph structure. We assume that the probability of a connection between nodes is higher when their features are more similar. To achieve this, we define a type-specific feature transformation layer for each node type $t \in \{H, L\}$

$$z_i^t = \sigma\left(z_i^t W_t^{\mathrm{FS}} + b_t^{\mathrm{FS}}\right) \tag{18}$$

where $W_t^{\mathrm{FS}} \in R^{F^{\mathrm{out}} \times F^{\mathrm{FS}}}$ is the transformation weight matrix, $b_t^{\mathrm{FS}} \in R^{1\times F^{\mathrm{FS}}}$ is the bias vector. Then, the cosine similarity kernel is employed to measure the similarity between node pairs of each relation $r \in \text{rel}$ as follows:

$$\phi_r^{\mathrm{FS}}\left(z_i^{r0}, z_j^{r1}\right) = \mathrm{sim}\left(z_i^{r0} \odot W_r^c, z_j^{r0} \odot W_r^c\right) \tag{19}$$

where $\odot$ is the Hadamard product and $W_r^c \in R^{1\times F^{\mathrm{FS}}}$ is the trainable weight vector that adjusts the contributions of different dimensions of features. $r0$ and $r1$ represent two types of nodes in the relationship r. For example, when $r = H \leftrightarrow L, r0 = H, r1 = L$. Then, the feature similarity adjacency matrix corresponding to each relationship r can be calculated as follows:

$$A_r^{\mathrm{FS}}[i,j] = \begin{cases} \phi_r^{\mathrm{FS}}\left(z_i^{r0}, z_j^{r1}\right), & \text{if} \phi_r^{\mathrm{FS}}\left(z_i^{r0}, z_j^{r1}\right) \geq \tau^{\mathrm{FS}} \\ 0, & \text{otherwise} \end{cases} \tag{20}$$

where $\tau^{\mathrm{FS}} \in (0, 1)$ is the threshold.

2) Neighborhood-Based GSL: Considering that similar nodes in the original graph may also have similar neighbors, we combine the similarity information obtained from the features of the dual-stream GAT with the first-order adjacency matrix to establish connections between a node and the neighbors of its similar nodes. This approach further enhances the feature propagation effect and provides more spatial context for similar nodes, addressing the issue of overly sparse node connections. Specifically, we compute a feature similarity matrix using the features output by the dual-stream GAT as follows:

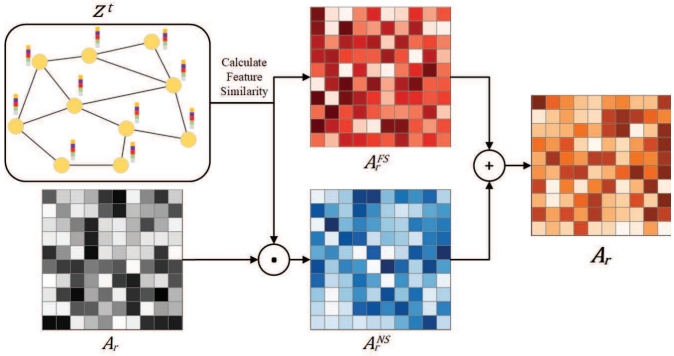


Fig. 7. Computing process of $\boldsymbol{A}_r$.

$$A_r^{\mathrm{FS}'}[i,j] = \begin{cases} \phi_r^{\mathrm{FS}'}\left(z_i^{r0}, z_j^{r1}\right), & \text{if } \phi_r^{\mathrm{FS}\prime}\left(z_i^{r0}, z_j^{r1}\right) \geq \tau^{\mathrm{FS}\prime} \\ 0, & \text{otherwise} \end{cases} \tag{21}$$

where $\tau^{\mathrm{FS}\prime} \in (0, 1)$ is the threshold. $\phi_r^{\mathrm{FS}\prime}$ is ϕ_r^{FS} with different weight vectors.

We define neighborhood similarity adjacency matrix as follows:

$$A_r^{\mathrm{NS}} = A_r^{\mathrm{FS}\prime} \odot A_r \tag{22}$$

where $A_r \in \{A^H, A^{H\leftrightarrow L}, A^L\}$. By computing in this way, feature similarity diffuses along the original graph structure and further generates a graph structure with a larger context. To obtain a symmetric adjacency matrix for similar neighborhoods, we refine the above equation as follows:

$$A_r^{\mathrm{NS}} = A_r^{\mathrm{FS}\prime} \odot A_r + A_r \odot A_r^{\mathrm{FS}\prime}. \tag{23}$$

3) Heterogeneous Graph Structure Fusion: Now, we have A_r, which contains the initial spatial adjacency relationships of the nodes, A_r^{FS}, which contains the similarity relationships between the nodes, and A_r^{NS}, which contains the neighborhood information. We employ a weighted strategy to integrate these different perspectives of the graph structure information and obtain the final learned optimized heterogeneous graph structure as follows:

$$\boldsymbol{A}_r = \sum_{1}^{3} A_i w_i \tag{24}$$

where $A_i \in \{A_r, A_r^{\mathrm{FS}}, A_r^{\mathrm{NS}}\}$ and $w_t \in R^1$ is the trainable parameter with $\sum_1^3 w_i = 1$. To maintain numerical stability, we regularize $\boldsymbol{A}_r$ as follows:

$$\boldsymbol{A}_r = \boldsymbol{D}^{-\frac{1}{2}} \boldsymbol{A}_r \boldsymbol{D}^{-\frac{1}{2}} \tag{25}$$

where $\boldsymbol{D}$ is the degree matrix of $\boldsymbol{A}_r$. The computing process of $\boldsymbol{A}_r$ is shown as Fig. 7. The $\boldsymbol{A}_r$ integrates three complementary perspectives: original spatial relationships, node feature similarity, and neighborhood structural consistency.

After obtaining $\boldsymbol{A}_r$, we can finally obtain the optimized heterogeneous graph structure as follows:

$$\boldsymbol{A}_{op} = \begin{bmatrix} \boldsymbol{A}^{H} & \boldsymbol{A}^{H\leftrightarrow L} \\ \boldsymbol{A}^{H\leftrightarrow L} & \boldsymbol{A}^{L} \end{bmatrix}. \tag{26}$$

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Algorithm 1 Process of Proposed DHGSLM

Input: Adjacency matrices A^H and A^L. Features Z^H, Z^L from dual-stream GAT.
Initialize the heterogeneous adjacency matrix A via formula (16).
for $t = H, L$ **do**
  for $i = 1, \ldots, n$ **do**
    Get $\mathbf{z}_i^t$ via formula (18).
  end for
end for
for $r = H \leftrightarrow H, L \leftrightarrow L, H \leftrightarrow L$ **do**
  Get feature similarity adjacency matrix A_r^{FS} via formula (20).
  Get neighborhood similarity adjacency matrix A_r^{NS} via formula (23).
  Get A_r via formula (25).
end for
Obtain optimized heterogeneous adjacency matrix A_{op} via formula (26).
Output: Optimized heterogeneous adjacency matrix A_{op}.

The process of DHGSLM is shown in Algorithm 1.

E. Loss Function and Classification

Effective feature representations of HSI and LiDAR have been derived through the dual-stream GAT and CLM, while the optimized heterogeneous graph structure has been learned via DHGSLM. To integrate the feature representations with the learned heterogeneous graph structure and achieve the final classification outcome, we define a straightforward GCN layer, as follows:

$$Y^{\text{pre}} = \boldsymbol{D}_{\boldsymbol{op}}^{-\frac{1}{2}} \left(\boldsymbol{A}_{\boldsymbol{op}} + \boldsymbol{I}\right) \boldsymbol{D}_{\boldsymbol{op}}^{-\frac{1}{2}} \left(Z^H \| Z^L\right) W^{\text{out}} \tag{27}$$

where $\boldsymbol{D}_{\boldsymbol{op}}$ is the degree matrix of $\boldsymbol{A}_{\boldsymbol{op}}$. Cross-entropy loss is employed to compute the classification error between the Y^{pre} and the labeled samples

$$L_{\text{CE}} = -\frac{1}{N_l} \sum_{i \in yl} \sum_{j=1}^{N_c} Y_{ij} \log Y_{ij}^{\text{pre}}. \tag{28}$$

The overall loss of the proposed HCGFN can be denoted as

$$\mathbf{L} = L_{\text{CE}} + L_{\text{BEM}}^H + L_{\text{BEM}}^L + L_{\text{LCL}} + L_{\text{NCL}}. \tag{29}$$

The process of HCGFN is shown in Algorithm 2.

Algorithm 2 Process of Proposed HCGFN

Input: An HSI I_H; An LiDAR I_L; label Y, number of iterations T.
Get HSI graph $G_H = (\mathcal{V}^H, A^H)$ and LiDAR graph $G_L = (\mathcal{V}^L, A^L)$ by input I_H and I_L into PAM.
for $t = 1, \ldots, T$ **do**
  Obtain Z^H and Z^L via formula (7).
  Calculate L_{BEM}^H and L_{BEM}^L via formula (9).
  Calculate L_{NCL} and L_{LCL} via formula (14) and (15) respectively.
  Obtain optimized heterogeneous graph $\mathbf{A_{op}}$ by input A^H, Z^H, A^L, Z^L into DHGSLM.
  Calculate classification result Y^{pre} via formula (27).
  Calculate L_{CE} via formula (28).
  Obtain overall loss $\mathbf{L}$ via formula (29).
  Update network parameters via operating back propagation.
end for
Output: Y^{pre} based on the trained network.

III. Experimental Results

In this section, the performance of our proposed HCGFN is evaluated on three public datasets, i.e., the Trento, the MUUFL, the Houston2013. Specifically, the three datasets are introduced in Section III-A. We introduce the experimental settings in Section III-B. In Section III-C, we analyze the hyperparameter settings of HCGFN. In Section III-D, we analyze the quantitative results of comparative experiments on three public datasets. In Section III-E, we explore the impact of the number of labeled samples on the accuracy. Finally, we analyze the results of ablation experiments to explore the effectiveness of each module of our method.

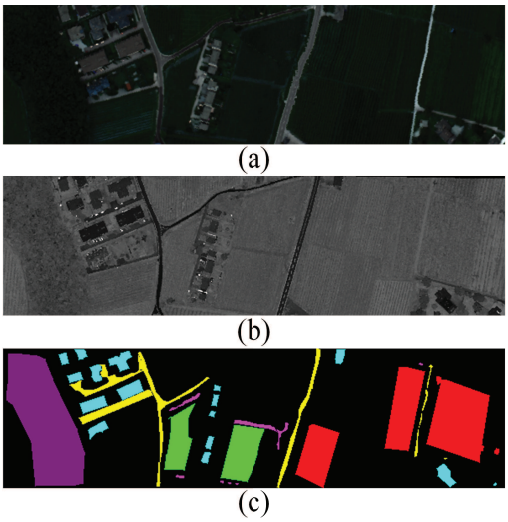


Fig. 8. Visualization of the Trento dataset. (a) False-color image of HSI. (b) LiDAR image. (c) Ground truth map.

TABLE I
Number of Training and Testing Samples for the Trento Dataset

ID	Color	Name	Train	Test
1		Apple trees	2	4032
2		Buildings	2	2901
3		Ground	2	477
4		Wood	2	9121
5		Vineyard	2	10488
6		Roads	2	3172

A. Datasets

1) Trento: The Trento dataset was collected at Trento, Italy. This dataset concludes an HSI and a LiDAR-derived DSM, both with 166 × 600 pixels at a spatial resolution of 1 m. The HSI data was captured by an AISA Eagle sensor with 63

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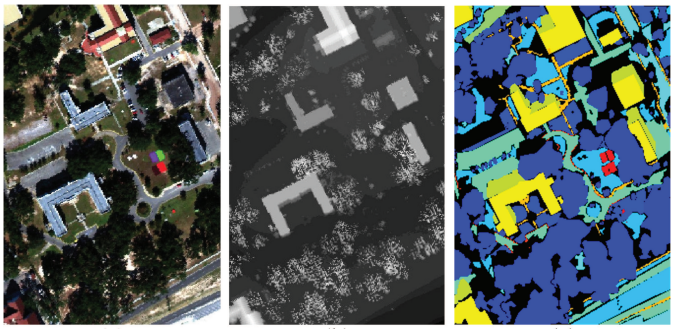


Fig. 9. Visualization of the MUUFL dataset. (a) False-color image of HSI. (b) LiDAR image. (c) Ground truth map.

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE MUUFL DATASET

ID	Color	Name	Train	Test
1		Trees	50	23196
2		Mostly grass	50	4220
3		Mixed ground surface	50	6832
4		Dirt and sand	50	1776
5		Road	50	6637
6		Water	50	416
7		Building Shadow	50	2183
8		Building	50	6190
9		Sidewalk	50	1335
10		Yellow curb	50	133
11		Cloth panels	50	219

spectral channels, which encompass a wavelength spectrum between 420 and 990 nm. The LiDAR data was captured by the Optech ALTM 3100EA sensor. This dataset includes six land-cover classes. Fig. 8 shows the false-color image, LiDAR image and ground truth map. Table I lists the labeled samples applied for training and testing for each class.

2) MUUFL: The MUUFL dataset [54] was collected in November 2010 at the University of Southern Mississippi Gulfport Campus, USA. This dataset concludes an HSI and a LiDAR-derived DSM, both with 325 × 220 pixels. The HSI data was captured by an ITRES CASI-1500 sensor with 72 spectral channels (64 available) at a spatial resolution of 0.54 × 1 m, which encompass a wavelength spectrum between 375 and 1050 nm. The LiDAR data was captured by the ALTM gemini sensor at a spatial resolution of 0.60 × 0.78 m. This dataset includes 11 land-cover classes. Fig. 9. shows the false-color image, LiDAR image and ground truth map. Table II lists the labeled samples applied for training and testing for each class. The MUUFL dataset comprises the following classes: trees, all tree species (especially live oaks) exhibiting varying canopy densities; mostly grass, ground visually dominated by grass; mixed ground surface, areas combining grass with soil, dirt, or other materials; dirt and sand, including dirt roads, bare soil surfaces, and bright white beach sand; road, covering both asphalt and gravel roads; water, all standing water bodies; shadow of buildings, the shadows cast by buildings may fall on asphalt or grass; buildings, all buildings; sidewalk, paved pedestrian walkways; yellow curb, yellow-painted curb edges marking boundaries; cloth panels, fabric panels with various spectral characteristics placed in the scene for experimentation.

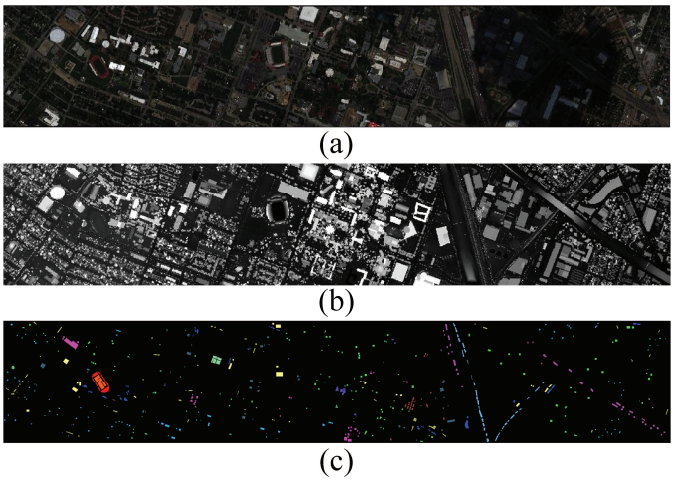


Fig. 10. Visualization of the Houston2013 dataset. (a) False-color image of HSI. (b) LiDAR image. (c) Ground truth map.

TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE HOUSTON2013 DATASET

ID	Color	Name	Train	Test
1		Health grass	50	1201
2		Stressed grass	50	1204
3		Synthetic grass	50	647
4		Trees	50	1194
5		Soil	50	1192
6		Water	50	275
7		Residential	50	1218
8		Commercial	50	1194
9		Road	50	1202
10		Highway	50	1177
11		Railway	50	1185
12		Parking lot 1	50	1183
13		Parking lot 2	50	419
14		Tennis court	50	378
15		Running track	50	610

3) Houston2013: The Houston2013 dataset was introduced in the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest, which was collected in June 2012 at the University of Houston, USA. This dataset concludes an HSI and a LiDAR-derived DSM, both with 349 × 1905 pixels at a spatial resolution of 2.5 m. The HSI data was captured by an ITRES CASI-1500 sensor with 144 spectral channels, which encompass a wavelength spectrum between 380 and 1050 nm. This dataset includes 15 land-cover classes. Fig. 10 shows the false-color image, LiDAR image and

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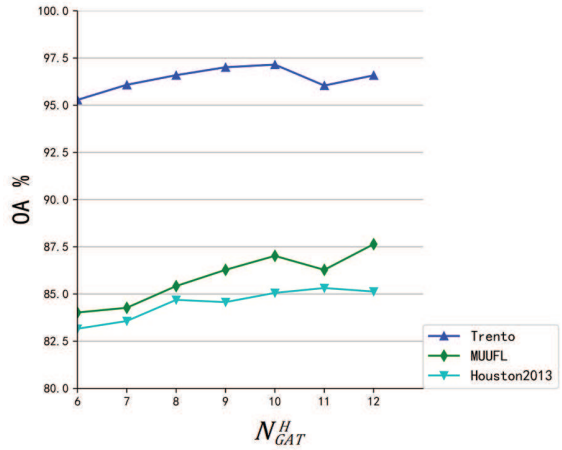


Fig. 11. Relationship between the OA and N^H_{GAT} on the three datasets.

ground truth map. Table III lists the labeled samples applied for training and testing for each class.

B. Experimental Settings

Our model is implemented by the open-source PyTorch 1.13.0 framework with Python 3.8.10. We set the learning rate as 0.0005 and the Adam optimizer is selected for updating the weights of the network. The training epoch is set as 1500. The hyperparameters are fine-tuned by the NNI 2.10.1 framework with a grid search strategy. To evaluate the classification performance of our HCGFN from multiple perspectives, we select ten baseline methods for comparative experiments. Specifically, our comparative analysis includes three conventional methods (KNN [55], SVM [56], and RF [57]), three state-of-the-art single-source classification methods (HybridSN [58], SpectralFormer [27], and S3-Net [59]), four state-of-the-art multisource methods (CCR-Net [11], FGCN [23], MFT [24], and GAMF [42]). The metrics include per-class accuracy, overall accuracy (OA), average accuracy (AA), kappa coefficient, and $f1$ score. The HSI and LiDAR are both used in all methods, HSI and LiDAR are concatenated band-wise for conventional and single-source methods. All experiments were conducted on a 64-bit Ubuntu server with a 24-GB GeForce 3090 GPU and Intel Xeon E5-2683 CPU with 32-GB memory.

C. Hyperparameter Analysis

1) Analysis for N^H_{GAT}: N^H_{GAT} controls the number of parallel GAT layers in the GAT encoder for HSI. To analyze the impacts of N^H_{GAT} on classification accuracy, we set N^H_{GAT} from the range of [6] and [12] with the interval of 1 for the three datasets. When N^H_{GAT} is set to a low value, the GAT encoder struggles to capture rich feature information, while a high N^H_{GAT} value may lead to overfitting. The evolution of OA along with N^H_{GAT} is shown in Fig. 11, we observed that when N^H_{GAT} is set to 10, 12, and 11, our proposed method achieves the best results on the Trento, MUUFL, and Houston2013 datasets, respectively.

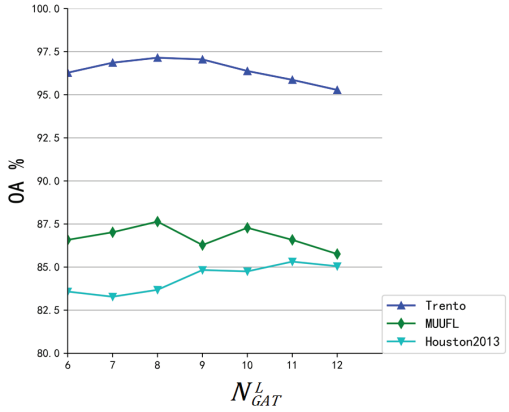


Fig. 12. Relationship between the OA and N^L_{GAT} on the three datasets.

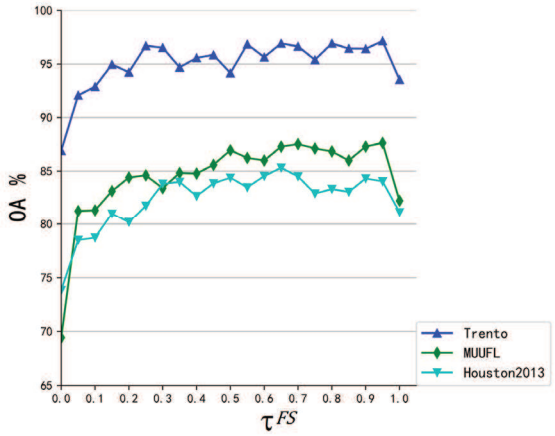


Fig. 13. Relationship between the OA and τ^{FS} on the three datasets.

2) Analysis for N^L_{GAT}: N^L_{GAT} controls the number of parallel GAT layers in the GAT encoder for LiDAR. To analyze the impacts of N^L_{GAT} on classification accuracy, we set N^L_{GAT} from the range of [6] and [12] with the interval of 1 for the three datasets. The evolution of OA along with N^H_{GAT} is shown in Fig. 12, we observed that when N^L_{GAT} is set to 8, 8, and 11, our proposed method achieves the best results on the Trento, MUUFL, and Houston2013 datasets, respectively.

3) Analysis for τ^{FS}: To analyze the impacts of τ^{FS} on classification accuracy, we set τ^{FS} from the range of $[0, 1]$ with the interval of 0.05 for the three datasets (see Fig. 14). τ^{FS} controls the sparsity of A^{FS}_r. When τ^{FS} equals 0, it means a FC graph is constructed, leading to all nodes being connected pairwise. When τ^{FS} equals 1, it means a disconnected graph is constructed, preventing the propagation of node information. The evolution of OA along with τ^{FS} is shown in Fig. 13. We observed that when τ^{FS} is set to 0.95, 0.95, and 0.65, our proposed method achieves the best results on the Trento, MUUFL, and Houston2013 datasets, respectively. The relatively smaller value of τ^{FS} for the Houston2013 dataset is due to its native graph structure being sparse, requiring more connections to enhance the features.

4) Analysis for τ^{FS}: To analyze the impacts of $\tau^{\text{FS}\prime}$ on classification accuracy, we set $\tau^{\text{FS}\prime}$ from the range of $[0, 1]$ with the interval of 0.05 for the three datasets. $\tau^{\text{FS}\prime}$ con-

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TABLE IV
CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE TRENTO DATASET (%) (TWO LABELED SAMPLES PER CLASS FOR TRAINING)

Class	KNN	SVM	RF	HybridSN	S3-Net	SpectralFormer	CCR-Net	MFT	FGCN	GAMF	HCGFN
1	**99.70**	68.85	49.68	70.21	96.80	43.30	87.33	94.52	64.43	98.66	93.43
2	**99.79**	90.69	47.64	46.67	48.47	88.49	86.49	97.00	98.07	95.14	89.98
3	59.12	97.69	86.79	76.10	52.62	51.36	74.00	92.03	69.81	33.33	**100.00**
4	65.72	96.46	53.54	**100.00**	45.20	76.87	95.91	98.75	**100.00**	**100.00**	99.44
5	0.00	69.79	51.77	71.65	49.25	97.95	98.41	95.67	**99.94**	88.58	97.75
6	0.00	86.54	42.06	23.80	85.94	67.56	24.15	46.53	76.01	52.43	**99.46**
OA	43.68	81.93	51.16	72.67	58.21	79.45	86.85	91.35	92.05	89.34	**97.15**
AA	54.06	85.00	55.24	64.74	63.05	70.92	77.71	87.42	84.71	78.02	**96.68**
Kappa	34.04	76.73	38.09	63.23	46.26	72.53	82.54	88.57	89.26	85.88	**96.21**
f1(×100)	33.84	75.06	46.04	61.94	57.70	75.32	71.77	79.85	83.45	78.70	**94.40**

TABLE V
CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE MUUFL DATASET (%) (50 LABELED SAMPLES PER CLASS FOR TRAINING)

Class	KNN	SVM	RF	HybridSN	S3-Net	SpectralFormer	CCR-Net	MFT	FGCN	GAMF	HCGFN
1	78.63	68.14	85.36	87.36	85.48	87.59	77.29	86.87	84.98	85.03	**90.13**
2	**87.23**	75.33	83.55	69.62	85.11	62.31	75.09	82.61	65.69	83.71	82.84
3	52.37	61.15	64.51	75.61	73.25	47.07	61.15	70.01	68.12	78.49	**81.22**
4	73.20	60.25	84.85	93.81	89.87	60.41	80.97	89.92	83.11	89.03	**96.57**
5	87.28	**88.87**	87.16	82.16	74.68	84.17	81.77	85.90	79.15	81.98	80.65
6	94.71	92.55	98.32	98.08	96.48	91.28	99.28	98.80	98.08	**99.77**	97.89
7	86.26	7.60	88.55	84.20	92.25	88.70	92.08	**94.27**	93.13	91.24	91.08
8	81.39	82.07	91.07	85.61	95.61	**95.88**	92.16	90.97	88.72	94.69	96.22
9	53.26	47.64	72.13	54.53	63.87	58.52	67.42	71.46	68.09	**75.76**	67.31
10	76.69	78.95	97.74	82.71	**100.00**	76.47	82.71	80.45	77.44	97.90	77.22
11	92.69	93.61	**99.54**	98.17	99.13	93.31	94.98	94.98	**99.54**	95.20	91.81
OA	77.01	69.08	83.40	82.96	83.78	79.33	78.08	84.85	80.98	85.18	**87.64**
AA	78.52	68.74	86.62	82.90	86.88	76.88	82.26	86.02	82.37	**88.44**	86.63
Kappa	70.92	61.92	78.67	78.03	79.25	73.06	72.39	80.56	75.64	81.03	**84.00**
f1(×100)	64.93	54.59	74.95	71.25	75.29	72.99	66.57	71.67	66.29	77.76	**77.99**

TABLE VI
CLASSIFICATION PERFORMANCE OF VARIOUS METHODS ON THE HOUSTON2013 DATASET (%) (50 LABELED SAMPLES PER CLASS FOR TRAINING)

Class	KNN	SVM	RF	HybridSN	S3-Net	SpectralFormer	CCR-Net	MFT	FGCN	GAMF	HCGFN
1	39.55	0.00	34.05	0.00	35.05	0.00	62.61	35.05	78.52	58.95	**83.44**
2	25.50	0.00	18.11	0.17	40.28	0.00	45.76	25.17	59.14	5.89	**75.08**
3	62.44	57.81	44.67	**100.00**	50.85	0.00	81.76	54.25	95.98	48.99	88.36
4	62.40	64.24	42.96	78.89	56.20	87.44	73.37	**93.22**	81.07	92.50	89.51
5	54.19	0.00	41.44	0.00	66.69	12.16	25.76	35.99	72.90	0.09	**81.18**
6	44.00	89.45	33.45	86.18	60.36	66.18	33.45	78.91	84.73	45.14	**97.97**
7	30.71	11.25	28.57	36.45	60.51	36.12	57.64	69.29	65.60	**93.02**	80.99
8	46.57	8.46	55.86	37.19	83.17	10.30	70.77	73.79	74.20	**89.21**	78.42
9	27.95	0.00	25.71	9.15	41.60	0.00	19.55	41.76	70.55	33.85	**98.45**
10	15.12	0.00	19.29	0.00	48.09	0.00	36.28	34.07	79.95	11.14	**83.40**
11	34.35	0.00	23.88	30.80	68.95	55.78	70.38	85.91	82.62	**95.02**	88.52
12	22.65	0.00	24.77	0.00	25.11	0.00	25.78	54.52	68.89	7.76	**78.42**
13	13.84	11.93	17.42	40.57	83.77	0.24	63.72	56.56	87.83	63.95	**92.74**
14	66.93	97.09	52.38	98.68	62.43	19.58	47.35	71.16	97.35	63.67	**100.00**
15	70.00	52.30	62.79	3.44	71.97	**100.00**	40.49	76.39	94.59	95.49	92.42
OA	38.90	16.53	33.59	26.30	54.65	22.97	50.06	56.71	76.52	51.38	**85.32**
AA	41.08	26.17	35.02	34.77	57.00	25.85	50.31	59.07	79.59	53.65	**87.26**
Kappa	34.26	13.10	28.50	22.40	51.14	18.14	46.18	53.50	74.64	47.49	**84.11**
f1(×100)	38.02	12.23	33.27	22.06	54.54	15.03	48.08	54.99	77.71	46.48	**87.82**

trols the sparsity of A_r^{NS}. As $\tau^{\mathrm{FS}\prime}$ decreases, the number of new neighbors that nodes can discover increases; as $\tau^{\mathrm{FS}\prime}$ increases, the amount of similarity information available for diffusion decreases. It can be observed that the optimal $\tau^{\mathrm{FS}\prime}$ is generally smaller than $\tau^{\mathrm{FS}\prime}$, indicating that the inclusion of more neighborhood information helps improve the feature representation.

D. Results

Tables IV–VII show the quantitative results obtained by different methods on three dataset, with the highest values in each row highlighted in bold. The corresponding visual results are illustrated in Figs. 15–20.

1) Results on the Trento Dataset: Nearly all methods achieved higher accuracy on the Trento dataset compared to the other two datasets. As shown in Table IV, among the traditional methods, SVM achieved higher OA, which can be attributed to its strong regularization capability, enabling it to find the optimal hyperplane in a high-dimensional feature space with fewer samples. In contrast, KNN, which relies solely on simple nearest-neighbor calculations, tends to overfit in situations where the sample size is small, as it overly depends on local features. Similarly, due to the limited number of samples, RF, when constructing decision trees, is susceptible to noise. An interesting observation is that SVM also outperformed the three single-source deep learning methods. These methods failed to consider and utilize the integrated complementary information between HSI and

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TABLE VII
OAs (%) OF THE PROPOSED METHOD BY DIFFERENT SETTINGS ON THREE DATASETS

datasets	PCM w/o LiDAR	without BEM	without CLM	without DHGSLM	HSI only	HCGFN
Trento	93.46	96.23	94.18	95.95	92.78	**97.15**
MUUFL	82.92	85.87	84.72	81.82	76.39	**87.64**
Houston2013	84.15	85.06	82.37	79.58	75.91	**85.32**

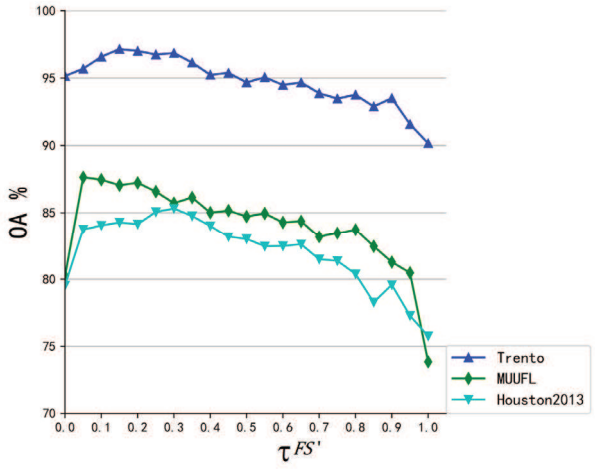


Fig. 14. Relationship between the OA and $\tau^{FS'}$ on the three datasets.

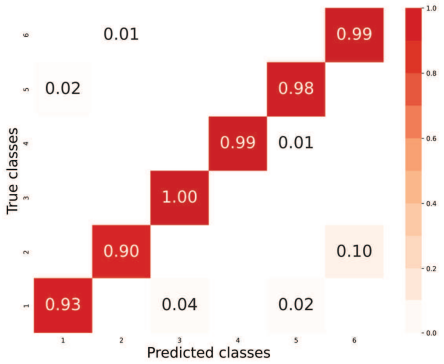


Fig. 15. Error matrix obtained by HCGFN on the Trento dataset.

LiDAR. In addition, in the extreme condition where there are only two training samples per class, deep learning methods cannot learn effective information, resulting in severe overfitting. In comparison, SVM can leverage the smaller sample size to learn an appropriately suited decision boundary when the data exhibits strong linear separability. However, SVM only maps the data features into a high-dimensional space to find the classification boundary. During this process, the spatial and geometric information of the data is lost, and the complementary information between HSI and LiDAR is ignored.

In contrast, multisource methods generally achieved better accuracy. Among them, CCR-Net employs a cross-channel reconstruction module and reconstruction strategy to facilitate the information interaction between HSI and LiDAR. However, its simple feature concatenation strategy does not fully capture the complementary information between the two modalities. MFT, using ViT as a feature encoder, achieves good performance by leveraging LiDAR to construct classification tokens for information complementarity and guidance. However, MFT does not consider the local positional information between tokens. GAMF utilizes CNN to extract tokens and applies GAT for information interaction between them. FGCN decomposes HSI and LiDAR into low-frequency and high-frequency components, and then fuses the corresponding components from both modalities, reducing information redundancy. Our proposed method, HCGFN, stands out as it simultaneously considers both the internal homogenous topological relations within HSI and LiDAR and the intermodal heterogeneous interactions, enabling full exploitation of their complementary information. Furthermore, compared to other methods, the proposed HCGFN model more effectively mitigates sample imbalance and achieves the highest $f1$ score. Moreover, the integration of BEM and CLM significantly enhances HCGFN's feature representation capacity under extreme sample conditions. Our HCGFN achieves the best OA. The error matrix obtained by HCGFN is shown in Fig. 15, it can be observed that our method performs well in each class.

As shown in Fig. 16, our method effectively recognizes long-distance objects, such as roads. In addition, compared with other methods, the increases of OA obtained by HCGFN are 53.48% (KNN), 15.23% (SVM), 45.99% (RF), 24.49% (HybridSN), 38.95% (S3-Net), 17.70% (SpectralFormer), 10.31% (CCR-Net), 5.80% (MFT), 5.10% (FGCN), and 7.82% (GAMF).

2) Results on the MUUFL Dataset: The MUUFL dataset contains complex scenes with numerous irregular objects and shadow interference. As shown in Table V, among the traditional methods, RF performs the best, while SVM struggles to correctly separate these complex categories using a simple maximum margin decision boundary. Under conditions of pronounced variation in test sample sizes across classes, the RF achieves a strong $f1$ score relative to other machine learning and single-source methods, owing to its inherent robustness to small sample sizes and class imbalance. KNN, due to its reliance on local distance metrics, is also hindered by the intricate interclass relationships. Unlike the Trento dataset, MUUFL has 50 training samples available for training for each class, which allows for a better performance of deep learning-based single-source methods. Among the single-source methods, S3-Net performs the best, as it employs a Siamese network for sample matching, enhancing interclass separability. SpectralFormer, however, performs poorly, possibly because its groupwise spectral embedding module does not effectively handle the HSI and LiDAR channels when constructing tokens. In the multisource methods, CCR-Net performs poorly, likely due to the fact that its simple feature fusion and reconstruction strategy is not well-suited

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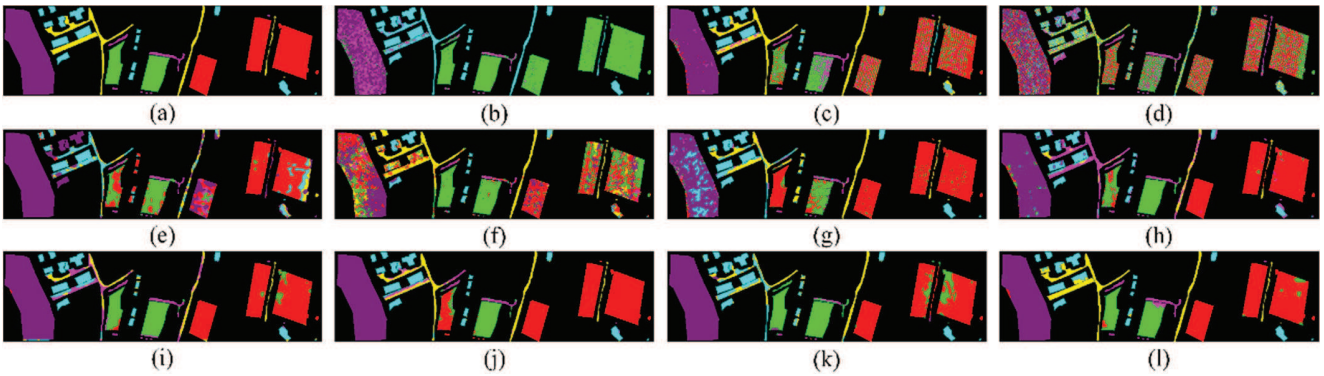


Fig. 16. Classification results of different methods on the Trento dataset. (a) Ground-truth map. (b) KNN. (c) SVM. (d) and (e) HybridSN. (f) S3-Net. (g) SpectralFormer. (h) CCR-Net. (i) MFT. (j) FGCN. (k) GAMF. (l) Our method.

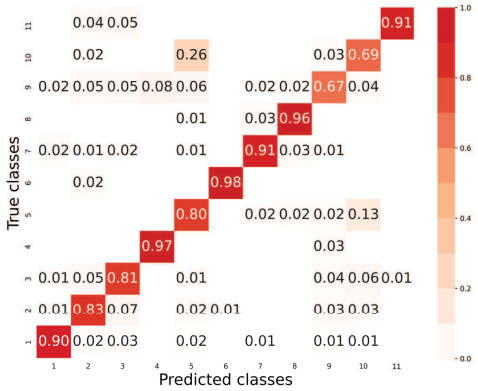


Fig. 17. Error matrix obtained by HCGFN on the MUUFL dataset.

for complex data. FGCN also underperforms, which may be due to its use of downsampling for low-frequency information, a strategy that may not be suitable for complex environmental features. As shown in Fig. 18, the results from FGCN contain many salt-and-pepper errors along the edges of large objects. GAMF, using CNN to extract contextual semantics for each pixel and employing GAT for token information interaction, achieves relatively better structure. However, GAMF fails to fully consider object-level semantics and relies solely on information from the training samples, neglecting the self-supervised information present in the unlabeled samples.

In contrast, our proposed HCGFN simultaneously incorporates both supervised and self-supervised information. In addition, through DHGSLM, we effectively extract and interact with complementary information, achieving the highest accuracy. The error matrix obtained by HCGFN is shown in Fig. 17, due to the similar spectral differences and elevation values between Road and Yellow curb, there are some misclassifications between them, but our method performs well in most classes. As shown in Table V, HCGFN achieves the highest OA and F1 score, owing to its ability to clearly distinguish buildings from their shadows and to accurately identify mixed ground surfaces. In addition, compared with other methods, the increases of OA obtained by HCGFN are 10.63% (KNN), 18.56% (SVM), 4.24% (RF), 4.68% (HybridSN), 3.86% (S3-Net), 8.31% (SpectralFormer), 9.56% (CCR-Net), 2.79% (MFT), 6.66% (FGCN), and 2.46% (GAMF).

3) Results on the Houston2013 Dataset: The Houston2013 dataset is characterized by sparse land cover, a large spatial extent, and a wide variety of land classes. As shown in Table VI, traditional methods perform poorly, primarily because many land classes in the Houston2013 dataset are similar, such as Parking lot 1 and Parking lot 2, making it difficult for these methods to identify simple linear classification boundaries. Among the deep learning-based single-source methods, S3-Net achieves higher accuracy, as it utilizes a Siamese network and metric learning to improve interclass separability. However, overall, these methods produce inconsistent classification results. As shown in Fig. 20, the inability to leverage the complementary information from HSI and LiDAR leads to difficulties in distinguishing spectrally similar land cover types, such as residential and commercial. In contrast, among the multisource methods, GAMF performs well in distinguishing between residential and commercial classes. However, as shown in Fig. 20, GAMF struggles to differentiate between highly similar land covers like stressed grass and healthy grass. This may indicate that, during the classification process, the balance between the influence of height and spectral information in the decision-making process is lost. FGCN, which uses fractional Gabor filters to extract variation information across different spatial scales and directions from HSI and LiDAR data, performs well in extracting classification boundaries for regular land covers such as synthetic grass and tennis court. However, FGCN struggles with small land parcels, as it can be influenced by neighboring land covers, which reduces classification accuracy, as seen in the case of stressed grass.

In comparison to the aforementioned methods, our proposed HCGFN achieves the highest accuracy. The fragmentation of land parcels in the Houston2013 dataset leads to an overly sparse initial graph. DHGSLM, however, effectively enhances information flow by constructing new global graph connections based on feature similarity, facilitating better interaction between homogeneous and heterogeneous data. HCGFN not only effectively distinguishes between highly similar yet spectrally different land covers, such as stressed grass and healthy grass, but also efficiently distinguishes spectrally similar yet height-differentiated land covers, such as parking lot 1 and parking lot 2. HCGFN concurrently achieves both high OA and high $f1$ score under severe class imbalance, demonstrating

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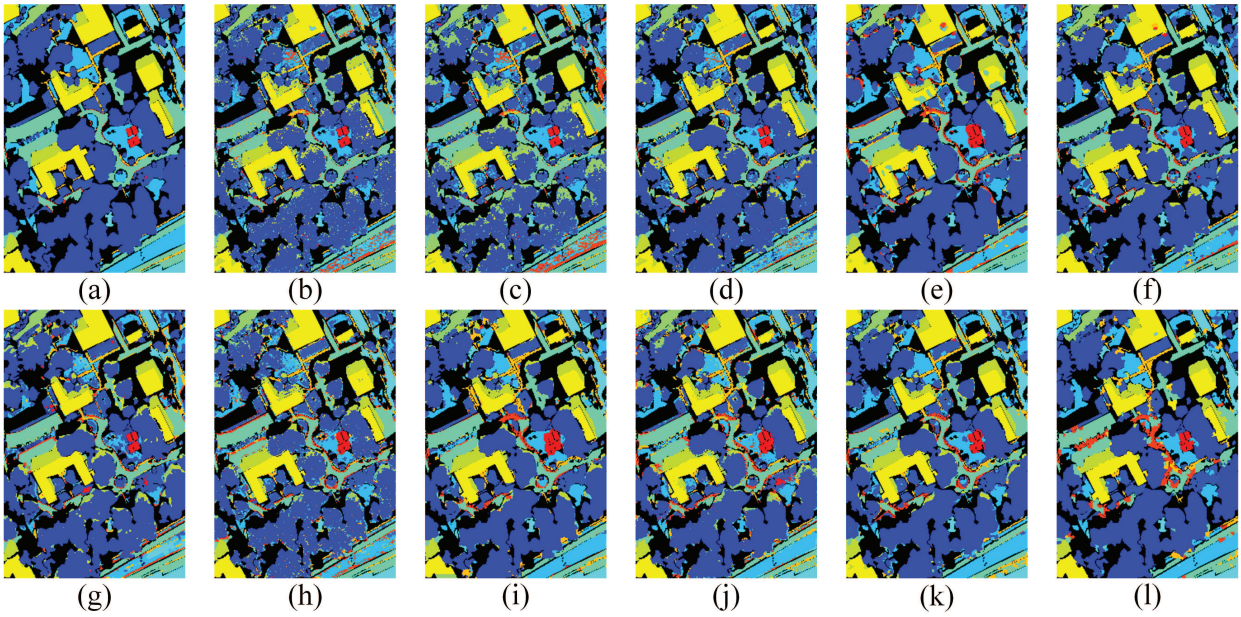


Fig. 18. Classification results of different methods on the MUUFL dataset. (a) Ground-truth map. (b) KNN. (c) SVM. (d) and (e) HybridSN. (f) S3-Net. (g) SpectralFormer. (h) CCR-Net. (i) MFT. (j) FGCN. (k) GAMF. (l) Our method.

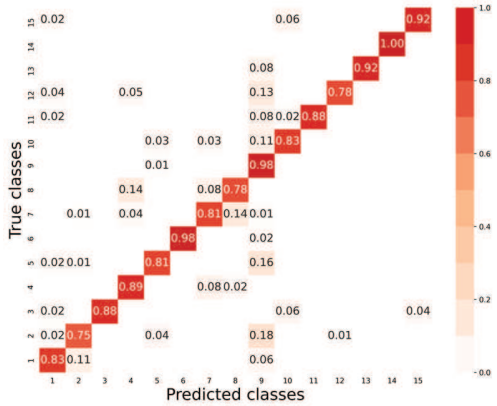


Fig. 19. Error matrix obtained by HCGFN on the Houston2013 dataset.

not only strong OA dominated by majority classes but also balanced precision and recall across minority classes. The error matrix obtained by HCGFN is shown in Fig. 19, some land features have been misclassified, such as road, highway, and railway, but our method maintains the highest classification accuracy on most classes. In addition, compared with other methods, the increases of OA obtained by HCGFN are 46.41% (KNN), 68.78% (SVM), 51.73% (RF), 59.02% (HybridSN), 30.67% (S3-Net), 62.35% (SpectralFormer), 35.26% (CCR-Net), 28.61% (MFT), 8.79% (FGCN), and 33.94% (GAMF).

E. Impact of the Number of Labeled Examples

In this section, we compare the performance of the proposed HCGFN with other methods under different labeled sample settings. For the Trento dataset, we gradually increased the number of labeled samples from 2 to 10. For the MUUFL and Houston2013 datasets, the number of labeled samples was gradually increased from 10 to 50. The experimental results, shown in Fig. 21, demonstrate that the classification accuracy of almost all methods improves as the number of labeled samples increases. As the labeled sample size grows, deep learning-based methods show a faster accuracy increase compared to machine learning-based methods, with multisource classification methods reaching higher accuracy limits. The proposed HCGFN consistently achieves the highest classification accuracy across all labeled sample configurations, which validates the capability of our method in effectively leveraging both supervised information from labeled samples and self-supervised information from unlabeled samples.

F. Ablation Experiments

As mentioned in Section I, our proposed MUUFL incorporates four critical components to enhance classification performance, i.e., PCM, BEM, CLM, and DHGSLM. For PCM, we simplified the input of PCM by designating configurations as "PCM w/o LiDAR" to indicate the removal of the LiDAR data. For other components, we simplified the model for analysis by designating configurations as "without BEM, " "without CLM " and "without DHGSLM " to indicate the removal of the BEM, CLM and DHGSLM components, respectively. Specifically, the "without DHGSLM" configuration replaces the $\boldsymbol{A}_{op}$ with the $\boldsymbol{A}$ for feature fusion.The configuration "HSI only" denotes the exclusion of BEM, CLM, and DHGSLM, and only using HSI for classification. The results of ablation experiments on three datasets are listed in Table VII. The results indicate that adding any component contributes to an improvement in classification accuracy, though the impact of each component on HCGFN's performance varies across different datasets. For PCM, the combination of LiDAR and HSI to generate superpixels yields higher OA than using HSI alone. For the Trento dataset, where the land cover is spatially well-distributed but the number of labeled samples is very small, CLM significantly enhances the model's performance under limited sample conditions.

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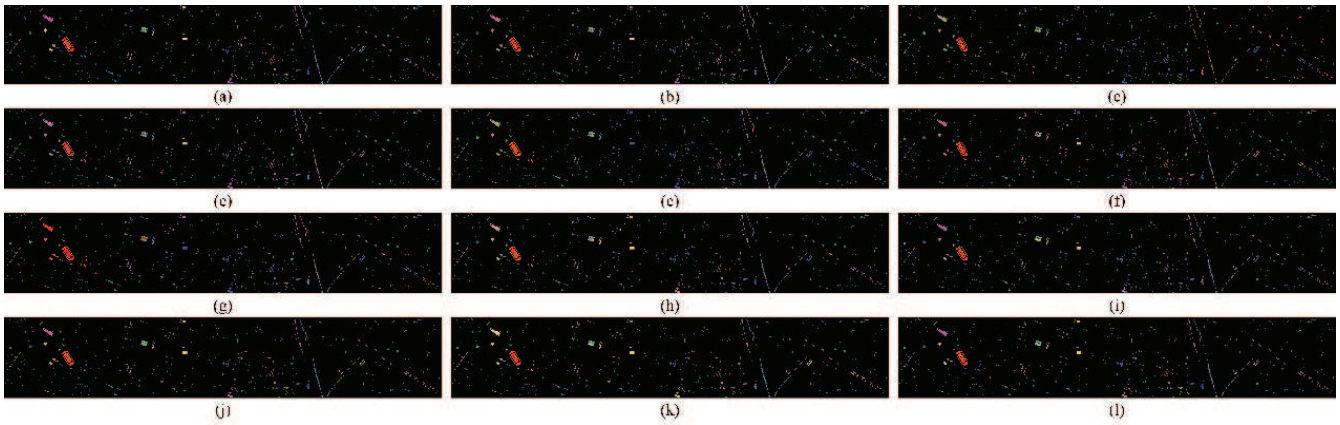


Fig. 20. Classification results of different methods on the Houston2013 dataset. (a) Ground-truth map. (b) KNN. (c) SVM. (d) and (e) HybridSN. (f) S3-Net. (g) SpectralFormer. (h) CCR-Net. (i) MFT. (j) FGCN. (k) GAMF. (l) Our method.

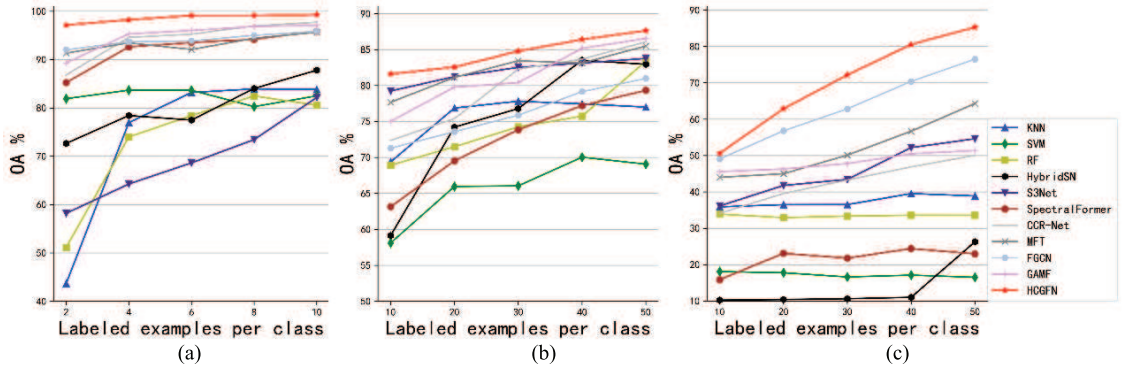


Fig. 21. OAs of various methods under different numbers of labeled examples per class. (a) Trento dataset. (b) MUUFL dataset. (c) Houston2013 dataset.

For datasets like MUUFL, which has dense and complex land cover, and Houston2013, where land cover is sparse and scattered, DHGSLM plays a crucial role by learning more meaningful heterogeneous graph structures, thereby contributing the most to the accuracy improvement.

IV. Conclusion

In this article, we propose the HCGFN method for classification of HSI and LiDAR data. In the proposed method, BEM is used to enhance the discriminability of modal features by utilizing label information. Furthermore, we propose the CLM to overcome the prevalent neglect of local topological self-supervision in existing methods while effectively integrating scarce labeled samples. Finally, to mitigate potential noise and incomplete connections in the initial graph topology, we propose the DHGSLM, which progressively refines both heterogeneous topology and node-specific fusion weights through dynamical similarity metric learning, thereby enabling effective cross-modal feature interaction and fusion. Extensive experiments on three datasets demonstrate that the proposed HCGFN achieves state-of-the-art results, with its internal components effectively boosting classification accuracy.

However, the proposed HCGFN remains limited to HSI and LiDAR data with the same spatial resolutions. Our future research will focus on developing resolution-adaptive fusion mechanisms to address cross-resolution multisource remote sensing feature fusion challenges. Furthermore, while the HCGFN demonstrates strong performance with limited labels, its dependency on supervised data persists. We plan to integrate a zero-shot learning paradigm with resolution-adaptive multisource remote sensing data classification.

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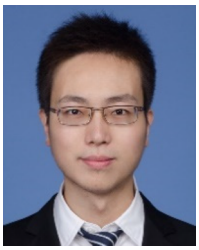
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